SURJECTIVITY OF THE RESTRICTION MAP IN GALOIS COHOMOLOGY FROM AN IMAGINARY QUADRATIC EXTENSION TO THE REAL PLACE FOR A CONNECTED SEMISIMPLE ALGEBRAIC GROUP OVER THE RATIONALS

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Editorial note. The counted unit is the author's labelled lemma KneserHasse at source lines 245-249 with the proof environment that begins two lines later at source line 251; the pairing is the ordinary adjacent one. The optional name of the lemma carries the citation to Kneser's lectures from which the statement is taken, whereas the proof printed here is the author's own argument, credited in the proof itself to A. Rapinchuk.

Editorial note. The retained counter values make the counted lemma print as 2.5, its number in the pinned source: the statement is the first numbered item of the excerpt and four proposition-class items precede it in source section 2.

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Allowed standard prerequisites (editorial).

- Galois cohomology of linear algebraic groups over a field of characteristic zero. Only the standard formalism of first Galois cohomology sets, restriction of scalars and the identification of the relevant classes of a torus with the elements satisfying t t^sigma = 1 is assumed. The construction that produces a representative defined over an imaginary quadratic field is the retained obligation.
- Weak approximation for tori at the infinite place. The density of the rational points of a rational torus in its real points is quoted as a standard arithmetic fact; every step that transports the cohomology class along that density is retained.
- Restriction of scalars for tori and the norm-one subtorus. The torus obtained from T by restriction of scalars from L to Q, the Q-automorphism tau and the subtorus of elements satisfying r r^tau = 1 are constructed inside the retained proof; nothing about them is assumed beyond the standard properties of restriction of scalars.

We begin by recalling a result of J. Tits that uses Galois cohomology to characterize the irreducible representations of semisimple algebraic groups over fields

Date: doi:10.3842/sigma.2015.034; CC BY 4.0.

that are not algebraically closed. However, we will state this work in the setting of Lie algebras, rather than algebraic groups. (We deal only with semisimple Lie algebras and semisimple groups, and the fields under consideration in this paper are all of characteristic zero, so no difficulties arise in making this translation [1, Proposition 7.3.1(iii), p. 393].)

Lemma 2.5 ([3, Theorem 5.1b, p. 77])**.** *If $\mathbf{G}$ is a connected, semisimple algebraic group over $\mathbb{Q}$, and L is any imaginary quadratic extension of $\mathbb{Q}$, then the natural restriction map $H^1\big(L/F; \mathbf{G}(L)\big) \to H^1(\mathbb{R}; \mathbf{G})$ is surjective.*

Proof. (The author thanks A. Rapinchuk for suggesting this argument.) Let σ be the nontrivial element of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$. It is well known that any cohomology class in $H^1(\mathbb{R}; \mathbf{G})$ is represented by an element t of a maximal torus $\mathbf{T}(\mathbb{C})$ of $\mathbf{G}(\mathbb{C})$, such that $tt^\sigma = 1$, and, since the variety of maximal tori has weak approximation, that $\mathbf{T}$ may be chosen to be defined over $\mathbb{Q}$ (cf. proof of [7, Proposition 6.17, p. 337]).

Let $\mathbf{R} = \mathrm{Res}_{L/\mathbb{Q}}\mathbf{T}$ be the torus obtained from $\mathbf{T}$ by restriction of scalars from L to $\mathbb{Q}$. Then there is a natural isomorphism $\varphi\colon \mathbf{T}(\mathbb{C}) \to \mathbf{R}(\mathbb{R})$, such that $\varphi\big(\mathbf{T}(L)\big) = \mathbf{R}(\mathbb{Q})$, and there is a $\mathbb{Q}$-automorphism τ of $\mathbf{R}$, such that $\varphi(x^\sigma) = \varphi(x)^\tau$, for all $x \in \mathbf{T}(\mathbb{C})$.

Let $\mathbf{R}^{(1)} = \{r \in \mathbf{R} \mid rr^\tau = 1\}$. We claim that this is a subtorus of $\mathbf{R}$. Since $\mathbf{T}$ is defined over $\mathbb{Q}$, there is an L-isomorphism $\psi\colon \mathbf{R} \to \mathbf{T} \times \mathbf{T}$, such that $\psi\big(\mathbf{R}(\mathbb{R})\big) = \{(x, x^\sigma) \mid x \in \mathbf{T}(\mathbb{C})\}$. Therefore, if we define $(x, y)^\pi = (y, x)$, then $\psi(r^\tau) = \psi(r)^\pi$ for all $r \in \mathbf{R}$. So $\psi(\mathbf{R}^{(1)}) = \{(x, x^{-1}) \mid x \in \mathbf{T}\}$ is a subtorus of $\mathbf{T} \times \mathbf{T}$, as claimed.

Now, $\mathbf{R}^{(1)}$ is a torus that is defined over $\mathbb{Q}$ (since the automorphism τ is defined over $\mathbb{Q}$), and all $\mathbb{Q}$-tori have weak approximation at the infinite place [7, Corollary 1 of Proposition 7.8, p. 418], so $\mathbf{R}^{(1)}(\mathbb{Q})$ is dense in $\mathbf{R}^{(1)}(\mathbb{R})$. Hence, some $q \in \mathbf{R}^{(1)}(\mathbb{Q})$ is in the same connected component of $\mathbf{R}^{(1)}(\mathbb{R})$ as $\varphi(t)$. Then, letting $q' = \varphi^{-1}(q)$, we see that q' is in the same connected component of $\{w \in \mathbf{T}(\mathbb{R}) \mid ww^\sigma = 1\}$ as t, so q' represents the same cohomology class as t in $H^1(\mathbb{R}; \mathbf{G})$. However, since $q' \in \mathbf{T}(L)$, we see that the cohomology class of q' is in the image of $H^1\big(L/F; \mathbf{G}(L)\big)$, as desired. □

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